Answers and talk notes

Reflect Points Across Axes

What your student is learning

Tonight we reflect a point across an axis in a real map problem — and check the distance from the mirror line.

Answers

Ask your student to explain before you look.

1. **A.** (4, 2). Over the y-axis, x changes sign: (4, 2).
2. **B.** 7 units. The x-coordinate measures distance from the y-axis, and |–7| = 7 units.
3. **C.** 2 units. The new point is (5, 2), and |2| = 2 units from the x-axis.
4. **D.** (9, 4). Over the y-axis, x changes sign, so the reflection is (9, 4), also 9 units from the axis.
5. **B.** (6, 1). Over the x-axis, y changes sign, so the decoy is at (6, 1).

Talk about it

How far is the chest from the mirror axis? How far is the decoy?

Home mission

Use a long table edge or a strip of tape as a mirror line. Place a coin 4 finger-widths from it. Can your student place the coin's mirror twin? Then move the coin closer. What happens to the twin?

When they are stuck, ask

- Which axis is the mirror line here?
- Which coordinate has the same name as that axis?
- Are both points the same distance from the mirror?

Try not to say

- Reflecting over the x-axis flips the x-coordinate.
- Change both signs for any reflection.
- The distance to the mirror can be different on each side.